Arjun **Jeewan**

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Separately applying formal methods and neural networks to biological questions has given me a grounded appreciation for what each can and cannot do. Neither is sufficient on its own, but the more interesting observation is that their limitations are complementary in ways that suggest a more integrated approach.
I want to do a PhD that explores that space seriously, bringing the full range of computational perspectives the question demands, and following it with the depth and rigor it deserves.

Research Experience

Advanced Centre for Treatment, Research and Education in Cancer (ACTREC) – Tata Memorial Centre — *Navi Mumbai, India*

Junior Research Fellow (on Project) — *July 2025 – Present*

- Developing **FuSION** (`github.com/supratik1/FuSION`), an open-source tool that identifies functionally significant regulatory interactions in noisy gene-regulatory networks by encoding KEGG pathway structure, STRING interaction-confidence scores, and microarray expression data as SMT constraints solved via Z3; engineered the C++ constraint encoding backend, achieving up to **4× speed-up** over the previous baseline (Akshay et al., CP 2019).
- Validating FuSION's inference against experimental microarray data and established GRN benchmarks (EGF signaling); under joint supervision of **Dr. Supratik Chakraborty** and **Dr. Akshay S.** (IIT Bombay) and **Dr. Prasanna V.** (ACTREC).
- Mapped heterodimer interfaces of five cancer-specific survivin isoforms with wild-type survivin via protein–protein docking, MD simulations, and MM-GBSA free-energy decomposition, identifying hotspot residues as candidate targets for isoform-selective therapeutic inhibition *(with Dr. Prasanna V.)*.
- Conducting structural analyses of DDX5/DDX17 RNA helicases and structure-based in silico inhibitor design targeting DDX17.

IHub-Data, International Institute of Information Technology, Hyderabad — *Hyderabad, India*

Healthcare AI Research Intern — *Sep 2024 – June 2025*

- Optimized the training pipeline of a standard Faster R-CNN (ResNet-50 + FPN) for abnormal cell detection on HMCHH-TCT-CellDet (Zhang et al., 2025). Training heuristic tuning produced **+4.8 AP_{50}** (67.4 to 72.2) over the prior best baseline with no architectural change and no added inference cost.
- Evaluated under 5-fold patient-wise cross-validation with COCO-style AP and FROC analysis; ablation isolates the contribution of each training heuristic.

Research Projects

Computational Drug Discovery for Type 2 Diabetes — *Goa, India*

BITS Pilani – with Dr. Raviprasad Aduri — *Jan 2023 – June 2023*

- Screened small molecules targeting the PPAR-γ transcription factor for insulin sensitization via a virtual screening → docking → MD validation pipeline; shortlisted **3 candidate compounds** with favorable predicted binding profiles.
- Performed molecular docking in Schrödinger Maestro (Glide) and confirmed candidate stability via Desmond MD simulations and GROMACS; top hits achieved docking scores competitive with reported PPAR-γ agonists in the literature.

RNA–Protein Interaction Prediction via Gradient Boosting — *Goa, India*

BITS Pilani – with Dr. Raviprasad Aduri — *Aug 2022 – Dec 2022*

- Extended the XRPI framework for RNA–protein interaction prediction by engineering structural features derived from high-resolution complex data, improving classification performance across diverse interaction types.
- Achieved **97.8% accuracy on NPInter** and **99.4% on TeloPIN**; validated via 10-fold nested cross-validation and held-out external datasets, with consistent performance across both lncRNA and mRNA interaction subsets.

Research Manuscripts

Jeewan A, Dholey M, Vinod PK. *Rethinking cervical cytology detection: A stronger Faster R-CNN baseline via training heuristics.* Manuscript under revision; preparing for resubmission.

Harish M, Mishra R, **Jeewan A**, Teni T, Venkatraman P. *In silico analysis of survivin heterodimers provides insights into the multifaceted role of survivin isoforms.* ACTREC – Tata Memorial Centre. Manuscript in preparation.

Presentations

Jeewan A. *FuSION: SMT-based identification of functionally significant regulators in noisy gene-regulatory networks.* Institute Seminar, ACTREC – Tata Memorial Centre, Navi Mumbai (forthcoming, June 2026).

Jeewan A, Dholey M, Vinod PK. *Rethinking cervical cytology detection: A stronger Faster R-CNN baseline via training heuristics.* Poster, R&D Showcase 2026, IIIT Hyderabad, March 2026.

Jeewan A, Dholey M, Vinod PK. *Cervic-AI: A Mamba state-space model for computer-aided diagnosis and abnormal cell localization in cervical cytology images.* Poster, R&D Showcase 2025, IIIT Hyderabad, March 2025.

Education

BITS Pilani, K.K. Birla Goa Campus — *Goa, India*

Bachelor of Engineering in Computer Science — *Oct 2020 – July 2024*

- CGPA: 7.81 / 10.0
- Relevant coursework: Machine Learning, Artificial Intelligence, Probability & Statistics, Theory of Computation, Data Structures & Algorithms, Database Systems, Object-Oriented Programming, Proteomics (ranked 1st in cohort)

Technical Skills

ML / DL	**Python, PyTorch, TensorFlow, Keras**, XGBoost, LightGBM, scikit-learn
Formal Methods	**C++, Z3 SMT Solver**, constraint encoding, SAT/SMT inference
Struct. Bioinfo.	**Schrödinger Maestro, AlphaFold 3, Rosetta**, GROMACS, Desmond, BioLuminate, AutoDock
Omics	**RNA-seq analysis, KEGG pathway integration**, GRN inference, MALDI-TOF peak analysis
Software Dev.	**Git, TypeScript, React**, MySQL, Postman

Test Scores

Nov 2023	**330 / 340**, Graduate Record Examinations (GRE) General Test	*Quant 168 · Verbal 162*
Nov 2023	**112 / 120**, Test of English as a Foreign Language (TOEFL iBT)	

Industry Experience

Jan–June 2024	**Software Engineering Intern**, AltiusHub	*Hyderabad, India*
June–July 2023	**NLP Engineering Intern**, New Street Technologies	*Bangalore, India*
June–July 2022	**Open Source Software Intern**, Swecha	*Hyderabad, India*

Service & Mentoring

BITS Goa Quiz Club — *BITS Pilani, Goa*

Co-Coordinator — *July 2022 – June 2023*

- ***Waves Quiz Fest*** (Nov 2022) — Organized 5 quizzes as part of BITS Goa's annual cultural festival, Waves.
- ***Brainstorm '22*** (Aug 2022) — Co-organized one of India's largest open online quiz fests; over 3,000 participants.

Peer Mentorship Programme — *BITS Pilani, Goa*

Peer Mentor — *July – Dec 2022*

- Mentored 4 junior students on coursework strategy, time management, and skill development.